EXP 7

Data Visualization Using Power BI

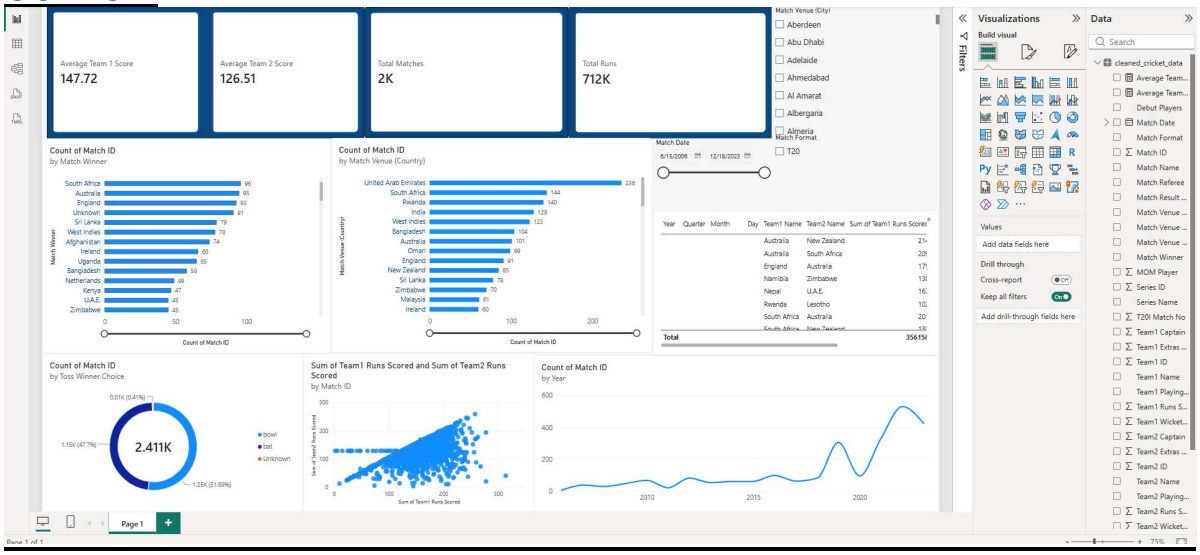
AIM:

To learn the Power BI Desktop workflow by connecting to a cleaned T20I cricket match dataset, cleaning data in Power Query, creating KPI cards and visualizations, and arranging them into an interactive dashboard to analyze cricket match performance.

ALGORITHM:

1. Open Power BI Desktop and identify the Visualizations, Data, and Filters panes, along with the Report, Data, and Model views.
2. Click Home → Get Data → Text/CSV, select the cleaned T20I cricket match dataset, and click Load.
3. Click Transform Data to open Power Query Editor and clean the dataset by handling missing values, removing duplicates where appropriate, and correcting data types.
4. Verify the columns, including Match ID, Match Date, Team1 Name, Team1 Runs Scored, Team2 Name, Team2 Runs Scored, Match Winner, Toss Winner Choice, and Match Venue (Country).
5. Click Close & Apply to load the cleaned cricket match data into Power BI.
6. Create four Card visuals displaying Total Matches, Total Runs, Average Team 1 Score, and Average Team 2 Score using suitable DAX measures.
7. Create a bar chart showing the number of matches won by each team using Match Winner and a distinct count of Match ID.
8. Create a line chart showing the yearly trend in T20I matches using Match Date and the distinct count of Match ID.
9. Create a pie chart to visualize toss decisions using Toss Winner Choice and the distinct count of Match ID.
10. Create a bar chart showing matches by country and a scatter chart comparing Team1 Runs Scored with Team2 Runs Scored for individual matches.
11. Arrange all KPI cards and charts neatly on one report page, add a suitable title, apply consistent colours and data labels, add slicers for Match Date, Match Venue (Country), and Match Format, and save the report as a .pbix file.

OUTPUT



RESULT:

Thus, the T20 International Cricket Match Performance Analytics Dashboard was created in Power BI to analyze total matches, total runs, average team scores, team wins, yearly match trends, toss decisions, country-wise matches, and team score comparisons using interactive visualizations.